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Dynamic Digital Twin-Driven Auto-Drilling: A New Era in Automated Well Construction

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Abstract

This article presents a cutting-edge approach to automated drilling, integrating an auto-driller system with advanced ecosystem powered by dynamic digital twin, to optimize drilling performance in real time. The methodology enables continuous adjustment of operational parameters based on actual wellbore conditions, significantly enhancing drilling efficiency, safety, and risk mitigation. By identifying operational sweet spots and automating critical processes, the system delivers substantial improvements in key performance indicators (KPIs). The paper includes implementation results and a comparative analysis, demonstrating the effectiveness of this intelligent ecosystem in modern well delivery.

The ecosystem provides functionality for pre-drilling planning and real-time execution phases. During the planning phase, a detailed drilling roadmap is created within the ecosystem, where operational parameters are validated through risk assessments and mechanical/hydraulic modeling to ensure wellbore and drillstring integrity. These verified parameters serve as target values for the auto-driller during execution.

In the operational phase, a dynamic digital twin continuously assesses wellbore conditions, calculates optimal operational parameters, and updates the roadmap accordingly. Real-time outputs—including setpoints for on- and off-bottom activities—are transmitted directly to the OEM Programmable Logic Controllers (PLCs), enabling fully unmanned and automated drilling. The system proactively detects the severity of drilling malfunctions and anticipates potential hazards, ensuring that key risks are mitigated at an early stage. Additionally, the auto-driller operates under a predefined control algorithm, which includes: automated operations sequence with continuous monitoring and adaptation based on current wellbore conditions. The drill-ahead module automatically defines the targeted toolface orientation for slide mode execution to follow the planned trajectory.

Implementing pre-drill planning within the ecosystem enables the selection of downhole equipment, identification of optimal drilling parameters, and estimation of expected ROP, taking into account the analysis of efficiency from previous offset wells drilling operations. Leveraging the ecosystem to control an auto-driller guarantees the execution of technological operations under optimal conditions. Wellbore, drilling equipment states are continuously and automatically evaluated in real-time by dynamic digital twin. Rock cutting process and Wellbore treatment efficiency, safe operational envelop and drilling dysfunction risks are automatically evaluated along the execution. Well construction cycle improvements exceed 30%

of the conventional drilling and predefined lookup auto-drilling techniques for similar wells. Using the drill ahead module has reduced sliding time by up to 10%. NPTs and ILTs losses were decreased by operating in the actual safe operational zone and mitigating at an early state downhole hazards such as: micro influxes and mud losses, pack-offs, stuck pipe, premature drilling tools wear, etc. and BHA overloads due to excessive torsional and lateral vibrations stick. The digital twin 1hz frequency automated analysis identified all precursors of upcoming hole condition deterioration: deviation of slack off weights and micro overpulls, automatically identifies FFs and generates sensitivity trend analysis, notifying key personnel, generating reports, and setting values to swiftly achieve optimal states.

The applied methodology proves its viability and effectiveness in well construction, particularly in settings with constrained operational pathways. Integrating the ecosystem that merges well design and real-time analysis with auto-driller systems presents an opportunity to deploy innovative process management techniques to achieve consistent improvements throughout available equipment. The integration of the digital ecosystem can seamlessly span across a diverse range of drilling systems, offering unprecedented flexibility to harness hybrid AI and physics-based insights.

Introduction

The oil and gas industry is undergoing a significant transformation, driven by the need for greater efficiency, reduced operational costs, and improved safety. One of the most promising developments in this context is the adoption of autonomous drilling systems (ADS) — an integration of hardware, software, and intelligent algorithms designed to perform and optimize drilling operations with minimal human intervention. These systems are at the forefront of the digital revolution in the upstream sector, enabling real-time decision-making, improving rate of penetration (ROP), reducing non-productive time (NPT), and enhancing wellbore quality.

One of the most promising frontiers for autonomous drilling systems is their integration with Industrial Internet of Things (IIoT) platforms and dynamic digital twins. Together, these technologies are redefining how drilling operations are designed, monitored, and optimized in real time.

While current autonomous drilling systems can execute closed-loop control and basic optimization tasks, their capabilities are limited by the scope of local data and pre-programmed logic. By connecting these systems to an IoT-based digital ecosystem and augmenting them with live digital twins, operators can unlock real-time cognitive capabilities across the entire drilling lifecycle.

IoT platforms enable seamless data exchange between surface equipment, downhole tools, logistics systems, and remote operational centers. This allows autonomous systems to make more holistic decisions based on the entire operational context — not just drilling mechanics. Unlike static models, dynamic digital twins are continuously updated using real-time sensor data from both the rig and the formation. These twins simulate current conditions and forecast future scenarios, enabling predictive adjustments to drilling parameters before issues arise.

Drilling automation technologies and stages overview

To better understand the evolution and current landscape of drilling automation, it is useful to examine the distinct stages of technological development—from traditional manual operations to fully autonomous, AI-driven systems.

1. **Manual Drilling**
   - **Description:** All drilling operations are performed manually by human operators.

- **Technology Level:** Minimal use of automation or monitoring systems. Operators control drilling parameters like weight on bit (WOB), rotary speed (RPM), and torque by direct manual input.
- **Challenges:** High reliance on operator skill, increased risk of human error, limited data collection, and less consistent drilling performance.

2. **Basic Automation**
   - **Description:** Introduction of automated control for fundamental drilling parameters such as weight on bit, rotary speed, and drilling torque.
   - **Technologies:**
     - Automated driller systems (autodrillers) that maintain preset WOB and RPM.
     - Basic sensors and alarms to monitor drilling parameters.
     - Partial automation of repetitive tasks (e.g., drill string handling cycles).
   - **Benefits:** Improves drilling efficiency and consistency compared to manual operation; reduces operator fatigue and manual intervention.
3. **Digital Rig 2.0**
   - **Description:** The rig integrates advanced digital systems to control and monitor drilling operations in real time. Automation is extended to multiple subsystems, supported by digital data acquisition and basic integration with management software.
   - **Technologies:**
     - SCADA (Supervisory Control and Data Acquisition) systems for process monitoring.
     - Real-time telemetry data streams from sensors across the rig.
     - Automated controls for mud pumps, top drive, drawworks, and downhole tools such as positive displacement motors (PDMs).
   - **Benefits:** Enables proactive decision-making, reduces non-productive time (NPT), improves safety via automated alarms, and supports better operational reporting.
4. **Digital Rig 3.0**
   - **Description:** This stage sees the emergence of advanced automation with AI-driven decision support and remote operations. Drilling cycles are highly automated, with real-time optimization based on machine learning algorithms and predictive analytics.
   - **Technologies:**
     - Artificial intelligence (AI) and machine learning models for optimizing drilling parameters dynamically.
     - Remote control and monitoring capabilities through cloud platforms.
     - Integration with Industrial Internet of Things (IIoT) devices for enhanced data gathering and control.
     - Automated wellbore steering and managed pressure drilling (MPD) systems with adaptive control.
   - **Benefits:** Significantly reduces human intervention, improves drilling efficiency and precision, enhances safety through predictive maintenance, and enables remote operation centers.

5. **Digital Rig 4.0 and Beyond**
    - **Description:** Represents a fully autonomous rig with robotic systems, digital twins, and immersive technologies. The rig can self-optimize and adapt to changing subsurface conditions with minimal human oversight.
    - **Technologies:**
        - Robotics for drill floor operations and tool handling.
        - Digital twin technology for virtual modeling and simulation of rig and wellbore behavior in real time.
        - Extended reality (XR), including virtual reality (VR) and augmented reality (AR), for operator training and remote assistance.
        - Self-learning automation systems with adaptive control algorithms continuously improving performance.
    - **Benefits:** Maximizes operational efficiency and safety, minimizes onsite personnel requirements, and enables continuous improvement through real-time simulations and feedback.

Table 1—Rig automation stages overview summary

Rig automation Stage	Key Technologies	Automation Level	Main Benefits
Manual Drilling	Manual control, basic sensors	0-10%	Operator skill driven; high variability
Basic Automation	Autodrillers, alarms, basic sensor feedback	20-40%	Improved consistency and operator relief
Digital Rig 2.0	SCADA, telemetry, real-time data acquisition	40-60%	Integrated monitoring and control
Digital Rig 3.0	AI, machine learning, IIoT, remote operation	70-90%	Predictive analytics, remote control
Digital Rig 4.0+	Robotics, digital twins, VR/AR, autonomous control	90-100%	Fully autonomous, continuous optimization

Digital ECO system overview

A unified IOT ecosystem was developed to provide a well construction full cycle support that integrates essential well construction stages: well design/pre-job phase, real-time operational support, autodriller systems control, job performance analysis.

This integration ensures seamless interconnectivity among the components, which can be configured to specific customer requirements. Developed toolkit complies with International Association of Drilling Contractors (IADC) [23] standards and addresses the needs of the oil and gas sector by incorporating a comprehensive data suite. The system is enhanced by artificial intelligence, which analyzes large volumes of data and identifies critical areas for improvement in well construction design and real-time decision-making.

A key aspect of this strategy is the flexibility to either consolidate well design, real-time support, and performance analysis within a single platform or to integrate functionalities from various suppliers. This approach aims to effectively leverage accumulated analytical data, facilitating strategic planning for future operations with advanced AI systems, thereby optimizing decision-making processes.

Deployment

Primarily, it is crucial to establish a continuous, real-time data collection from drilling operations. Two viable options are available: 1. Deploying EDGE device (Gateway) on the drilling rigs. 2. Connecting to a WITSML or OPC server to ensure real-time access to real-time data remotely.

Each option presents unique advantages and challenges. The advantages of EDGE devices include: 1. Enhanced capabilities for managing an Auto-driller PLC. 2. Immediate alerts on technological deviations to rig site personnel. 3. Fully functional autonomous in case of network issues. The advantage of using a remote data server is a Simplified connectivity while the other side of the coin is potential scalability and data quality risks and internet connectivity dependencies.

The diagram on the Fig.1 illustrates the integration of physical components into a digital platform designed for comprehensive well construction management. It consists of four main elements: the Field, Data Center – Cloud, Remote Monitoring, and Remote Control Center. In the field, the EDGE device collects real-time sensor data, performs digital twin analyses, visualize and set operational objectives to personnel or directly to autodriller PLC. All gathered data undergoes quality control checks and then transmitted to the Data Center. The Data Center serves as a central hub, processing this data from all drilling rigs and performing advanced multiwell analytics to provide actionable results. Remote Monitoring allows clients to oversee rig operations. The Remote Control Center enables engineers and managers to make informed decisions and remotely control drilling parameters based on actual data. Data is transferring continuously from the drilling rigs to the Data Center and back, forming a feedback loop that ensures continuous optimization of drilling operations. This integration facilitates real-time monitoring, operational support, and performance analysis, leveraging advanced analytics and AI to improve decision-making and operational efficiency.

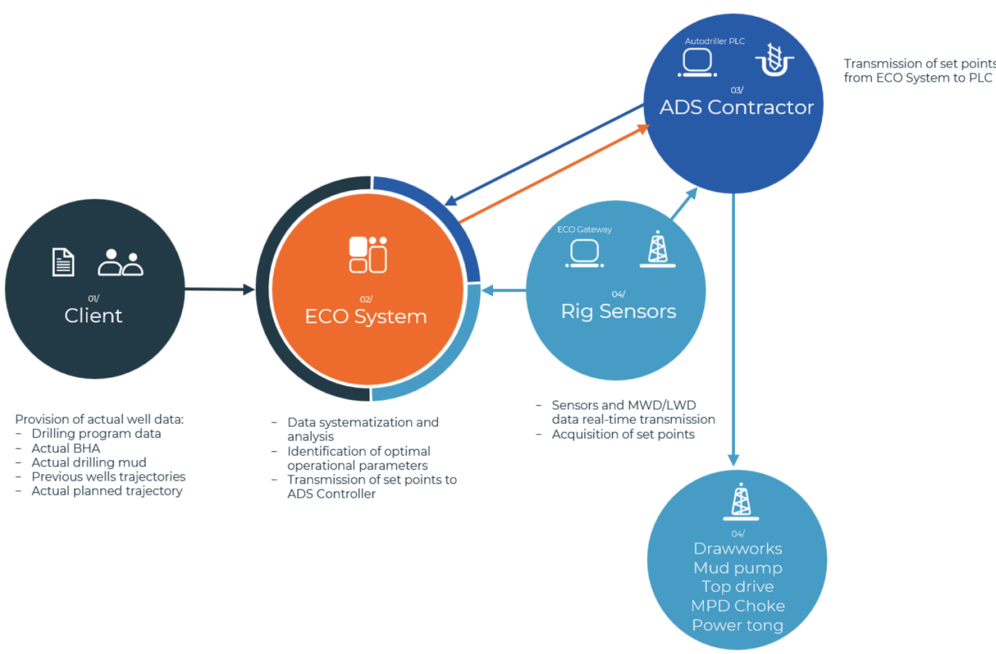

Figure 1—System EDGE devices-based deployment solution.

Framework description

The functional diagram the proposed framework is shown on the Figure 2.

The diagram illustrates comprehensive system for managing well construction, integrating various components such as well design, real-time monitoring, real-time operations, job performance analysis, and AI systems. In the well design phase, static well data is collected and processed to create detailed drilling plans and geological hazard roadmaps based on AI analytics data results. Real-time monitoring involves the use of sensors and digital twins to collect and analyze drilling data, supported by tools like smart alarms, real time anticollision analysis, automated broomstick plots, etc. Real-time operations are managed through automated driller systems and operational roadmap. Job performance analysis utilizes advanced data analytics for gathering important drilling data, calculate key performance indicators and detect non-productive time automatically. Analytics AI-based data acquisition and processing systems analyze large volumes of data and identify critical areas for improvement in well construction design and real-time decision-making.

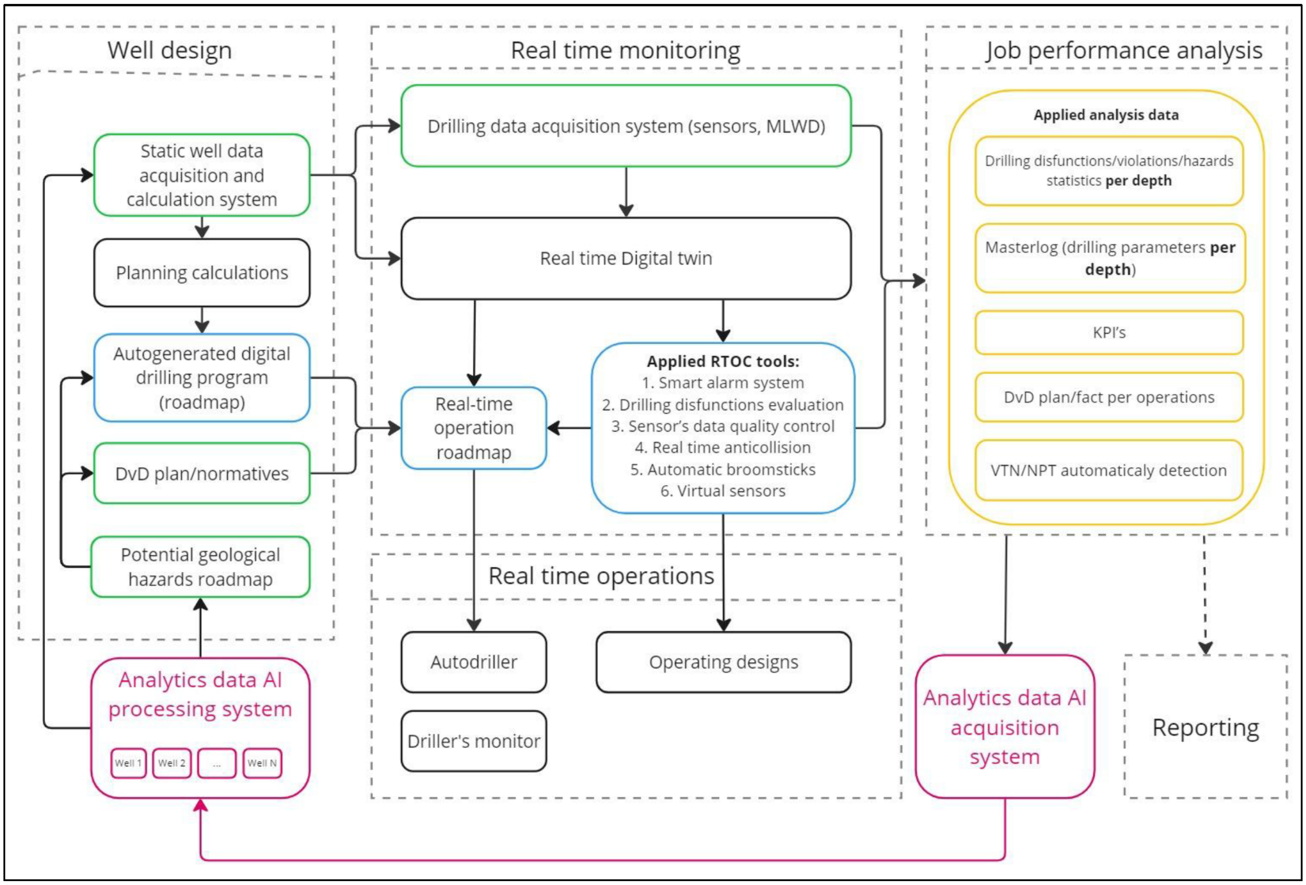

Figure 2—Functional schema of proposed approach.

Well Design

The outcome of this module is a comprehensive well design. A mandatory requirement for this module is the ability to historical drilling data to optimize the design. The module consists of the following components:

1. Well data and planning calculations as a result.
2. Autogenerated digital drilling program - a roadmap for autodriller execution.
3. Depth vs Day schedule and Potential geological hazards roadmap.

    1. *Well Data.* The primary output of this module is the design of well using accumulated historical drilling data. This includes well section plan, well trajectory, anticollision analysis, mechanics/hydraulics calculations, selection of drill string and BHA components, BHA vibration analysis, and selection of drilling fluids properties.
    2. *The autogenerated digital drilling program* is Planning Drilling Operation Roadmap that automatically assesses all technological limitations based on the initial data determined at step 1 in the Static well data acquisition/calculation system. Then the optimal operational parameters define for ROP maximizations considering potential geological hazards. The resulting operational parameters are compared with the optimal operational parameters identified through the analysis of accumulated historical data using advanced AI analytics.

Importantly, it provides recommendations for changes to input data such as BHA, fluids, trajectory, sections, etc. A key feature of the autogenerated digital program is the ability to provide feedback message regarding the calculated initial data and identify sweet spots for autodriller system. These modules are fully integrated and complement each other. A very important result of AI analytics is the potential geological hazards roadmap, which is used to highlight specific intervals in the Drilling Operation Roadmap and operational parameters clarifications within them.

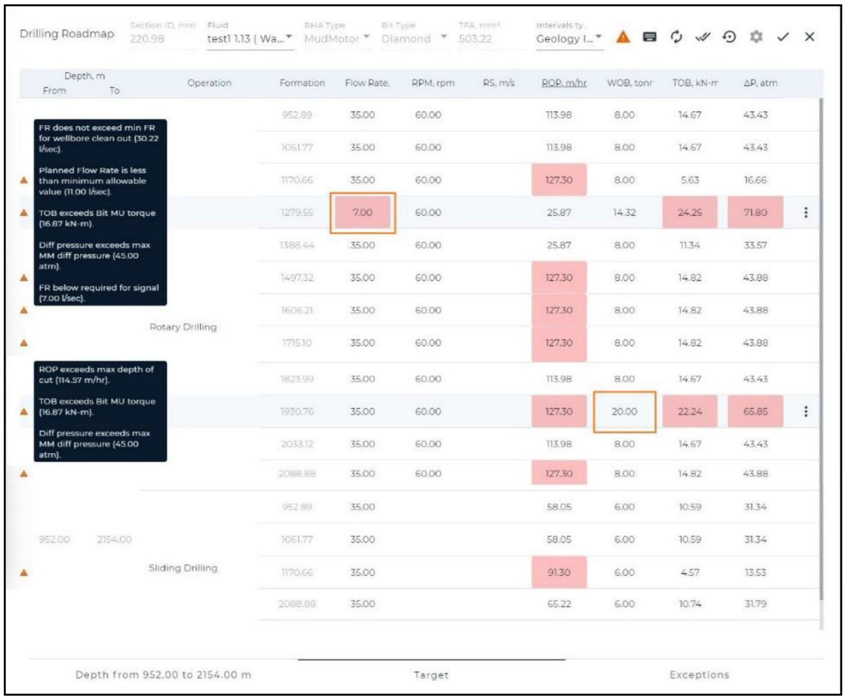

Figure 3—Planning Drilling Operation Roadmap. Feedback messages in the dark fields because of intentional manual input of wrong flow rate (7 l/s) and WOB (20 t) parameters for demonstration purposes.

Real-time operational support

This module provides extensive operational support for drilling, encompassing two critical aspects: 1. Real-time monitoring, and 2. Real-time operations. The real-time monitoring component incorporates a suite of tools deployed at the Real-time Operations Control Centers (RTOC). These tools facilitate essential calculations and analytics during the drilling process. The primary benefit of employing these tools is equipping the drilling engineers at the RTOC with comprehensive analytics crucial for informed decision-making and the effective management of real-time operations, that Real-time operations module represents. A foundational element of this framework is the integration with the digital well-design, which was created during the initial stage, ensuring that operations are aligned with the planned specifications and enhancing overall operational efficiency. [R. Karpov et al., 2024]

Applied RTOC tools:

1. Real-time Digital Twin: Executes real-time multifaced analysis being fed live instrumentation data to evaluate current state and lookahead.
2. Smart Alarm System: Evaluates operational parameters regimes deviations from the Digital Twin's calculations. Customizable calculations are available.
3. Drilling Dysfunction Evaluation/Trend Analysis: Analyzes patterns and potential issues while drilling operations.

4. Sensor Data Quality Control: Ensures the sensor data reliability and accuracy control.
5. Real-time Anticollision: Evaluate the collision risks with neighboring wellbores.
6. Automatic Broomsticks (Friction Factors Trend Control)
7. Automated Slide Sheet
8. Virtual Sensors
9. Downhole Mechanical Specific Energy (MSE) real time calculations
10. Real-time Operational Roadmap/Optimization of Rate of Penetration (ROP): Guides well construction operations including rotary drilling, slide drilling, reaming, back reaming, tripping in/out in automated mode to optimize efficiency and effectiveness. The ecosystem toolkit can be customizable to meet the specific needs of each client and is not confined to the components listed above. This toolkit is actively utilized by our clients to enhance operational efficiency and safety in drilling operations.

    1. ***Real-time Digital Twin technology*** implements mathematical models that describe the physics of downhole processes and evaluates the well construction process in real time, every second. The Digital Twin leverages data and physics insights, considering coupled well hydraulics, geomechanics, well thermodynamics, work string mechanics & drill bit rock interaction effects. This dynamic twin predicts expected values and evaluates the accuracy of current sensors' readings in real time. The methodology becomes an effective tool to assess sensor data quality online, and it allows timely service instrumentation to resolve raised sensor issues, filter outliers, and switch to alternative sensor readings when available (Karpov et al., 2021).
    2. ***Smart Alarm System:*** In conjunction with the Dynamic Digital Twin, the smart alarm system operates with dynamically determined threshold values based on Digital Twin calculations as well as manually defined static thresholds. When deviations occur, the system triggers alerts and records events in the database. The alarm system can be customized for any operational parameters, which are categorized into operational, wellbore quality and integrity, work string, and surface equipment:

        - *Operational thresholds* – limits characterizing violations of operational standard procedures, drilling plan roadmap, regulations, and established best practices (e.g., hook load, standpipe pressure, surface torque, running speed tripping in/out, flow rate, RPM, inactive time in open hole, weight on bit, pressure drop on mud motor, etc.).
        - *Wellbore thresholds* – limits violation of which exposes a risk to wellbore quality and integrity (e.g., ECD, friction factors during rotation, tripping in/out, hole cleaning index, pits volume, etc.).
        - *Work string thresholds* – limits the violation of which compromise the integrity of the drilling string and BHA components.
        - *Surface equipment thresholds* – limits identified by safety margins and specifications of surface equipment.

        The outcomes of the Smart Alarm System include:

        - Instant notifications for rig personnel via the drillers' HMI, messenger or email
        - Statistics on deviations, which are collected in an Analytics Data AI Acquisition System.

    3. ***Drilling dysfunction*** module evaluates in real time risks such as cleanout, twist-off, drill string failure, influx, mud loss, washout, and helical buckling risks. Statistics on these risks are also gathered in the Analytics Data AI Acquisition System.
    4. ***Sensor's Data Quality Control*** [R. Karpov et al., 2023] minimizes the amount of incorrect data for real-time system operation subsequent analysis, achieved through the following means:

- Determination of invalid sensor data (absence of value/signal, high/low/static values), ability to select a valid data source in an automatic regime.
- Data QC against alternative data sources.
- Data QC against coupled hydromechanical model – Real time digital twin.

5. ***Real-Time Anticollision Tool***. This system tool is pulling directly from MLWD unit and calculates collision risks of with neighbor wells. The modules features:

   - Alerts and Notifications: Provides real-time alerts in case of collision risk.
   - 3D Visualization: Displays high well collision risk intervals for enhanced spatial understanding.
   - Ellipse of Uncertainty Flexible Settings: Allows adjustment of uncertainty parameters to improve risk assessments.
   - Anticollision Analysis Report: Summarizes findings and recommendations for avoiding collisions.

6. ***Automated Slide Sheet.*** This tool aggregates drilling modes data, distinguishing between slide and rotary drilling activities. It visualizes MWD surveys and calculated friction factors by intervals, reducing manual inputs required from the DD (Directional Driller) engineer during well construction. The system automatically creates a table displaying drilled intervals and includes average drilling metrics such as RPM, ROP, flow rate, pressure off bottom, torque, hook load (up, down, rotation), and survey including MD, TVD, inclination, azimuth, and DLS. Slide sheet log is available in Excel report format.
7. ***Virtual downhole Sensors.*** Mathematical modeling of well construction operations has proven to be a reliable tool being able to replace a need for a physical sensor with a virtual-soft sensor (Gumus et al., 2014), (van der Geest et al., 2001). The soft sensor approach has been successfully used to enhance the accuracy of derived drilling parameters (Kyllingstad et al., 2019), (Tikhonov et al., 2014) and to determining abnormal readings measured by sensors (Cayeux et al., 2014). The system calculates in real time the following parameters: effective WOB, TOB, bit RPM, ECD on bottom, bit and casing shoe.
8. ***Mechanical Specific Energy (MSE) calculations*** profoundly improve real-time decision-making and troubleshooting throughout drilling processes. Suggested methodology integrates downhole MSE, made possible by Virtual sensors through a Digital Twin. Downhole MSE typically yields more precise data regarding the actual conditions encountered downhole, thereby enhancing the accuracy and effectiveness of drilling operations:

   - Equipment Failure Prevention: By monitoring MSE in real-time, the platform can detect abnormal spikes that might indicate impending failures in drill bits or other downhole equipment. This proactive approach allows for timely interventions, preventing costly downtime and enhancing safety.
     Implementing safety MSE limits in Smart Alarm system allows to control MSE automatically.
   - Enhanced Wellbore Stability: MSE data provides insights into the strength of the formation, aiding in decisions that enhance wellbore stability. For example, a low MSE value might indicate low formation strength, which can guide adjustments in drilling parameters and additional reaming to prevent wellbore collapse.
   - Bit Selection and BHA Design: The integration of MSE with a well-placement methodology includes optimized bottom hole assembly (BHA) design and bit selection based on historical drilling data. This optimization aims to maximize drilling efficiency and accuracy.

- Real-Time Geosteering [24]: Using MSE along with downhole drilling dynamics data from a drilling optimization collar allows for precise real-time geosteering. This system helps maintain the planned trajectory within a thin target layer while minimizing borehole tortuosity and accurately navigating geological challenges.
- Stratigraphic Correlation and Porosity Boundaries [24]: MSE data correlates well with real-time logging-while-drilling data, aiding in quick decision-making regarding stratigraphic shifts or approaching different porosity boundaries. This capability helps reduce undesired borehole azimuth swings and ensures the drill path stays on target.

9. ***Real-time Operational Roadmap/Optimization of Rate of Penetration (ROP)*** The Real-time operational roadmap facilitates the selection of optimal drilling parameters, considering downhole and surface equipment specifications as well as rock formation properties. This methodology allows for the precise configuration of drilling parameters such as the Rate of Penetration (ROP) or Weight on Bit (WOB), and the computation of expected surface torque, Torque on Bit (TOB), bit RPM, mud motor pressure drop, minimum required flow rate, maximum allowable surface torque, and the maximum WOB allowable before helical buckling occurs.

This Real-time drilling roadmap also ensures that the selected operational parameters are continuously validated against the equipment capabilities and the geological conditions encountered. It alerts users immediately if any parameters or specifications are breached, thus maintaining drilling integrity and efficiency. Additionally, the operational roadmap segments parameter selection by rock formation intervals or stand length, optimizing operations based on the specific geological context. [R. Karpov et al., 2024]

The real-time operational drilling roadmap can be integrated with the auto-driller system PLC, ensuring seamless rig activities automation performance.

Real-time drilling roadmap execution description

The ecosystem is designed with high interoperability and is capable of interacting with PLCs from various manufacturers. Its flexible architecture ensures seamless integration regardless of the specific hardware in use.

Built-in algorithms support multiple levels of drilling automation, from Stage 2 to Stage 4, depending on the autodriller's capabilities, the system configuration, and the overall automation level of the rig.

When integrated with an advanced autodriller capable of managing pumps, drawworks, and the top drive, the ecosystem enables fully automated slips-to-slips cycles. Drilling, reaming, and tripping operations can be executed automatically without driller intervention. While connection operations are still performed manually by the rig crew, the system resumes automated control immediately after each connection, ensuring continuous and optimized execution of the drilling roadmap.

During the operational phase, the execution of the drilling roadmap is critical to achieving efficiency and wellbore quality. Integration with the auto-driller's PLC allows for the seamless transfer of setpoint values derived from the roadmap, enabling precise control over drilling parameters.

The ecosystem operates in real time, leveraging a dynamic digital twin and advanced analytical tools to continuously monitor and adjust operational regimes based on data from surface sensors and the MLWD (Measurement While Drilling) unit.

Throughout the drilling process, the system constantly evaluates and adjusts the roadmap using high-frequency digital twin computations. These calculations consider the actual wellbore trajectory, friction factors, BHA configuration, specifications of both downhole and surface equipment, drilling fluid properties, and geological formation characteristics. Based on this evolving information, the system identifies the optimal operating parameters—so-called "sweet spots"—such as Weight on Bit (WOB), Rate of Penetration (ROP), Rotations Per Minute (RPM), and flow rate for each formation layer and drilling mode, both rotary and slide.

The Drilling Roadmap defines both the planned and maximum allowable values for key parameters such as WOB, ROP, RPM, and flow rate. (Fig. 4, 5). These values are configured specifically for each rock formation and drilling mode. For every parameter, a target value and a corresponding operational limit are established—for example, WOB and Max WOB, or ROP and Max ROP.

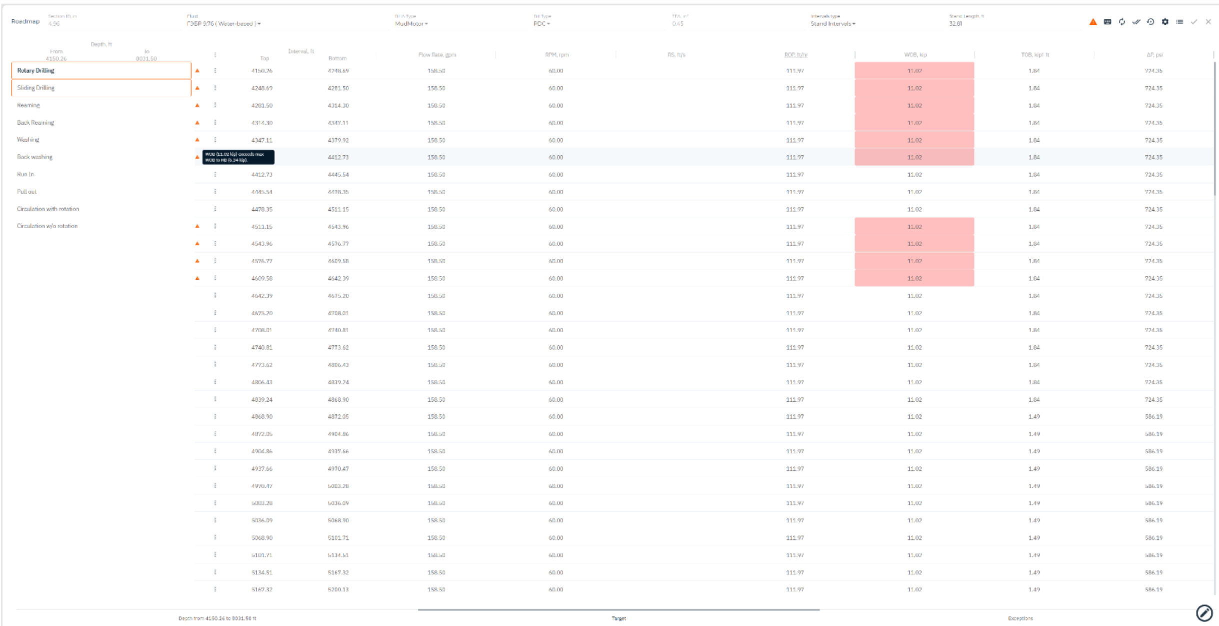

Figure 4—Drilling Road map in the ecosystem

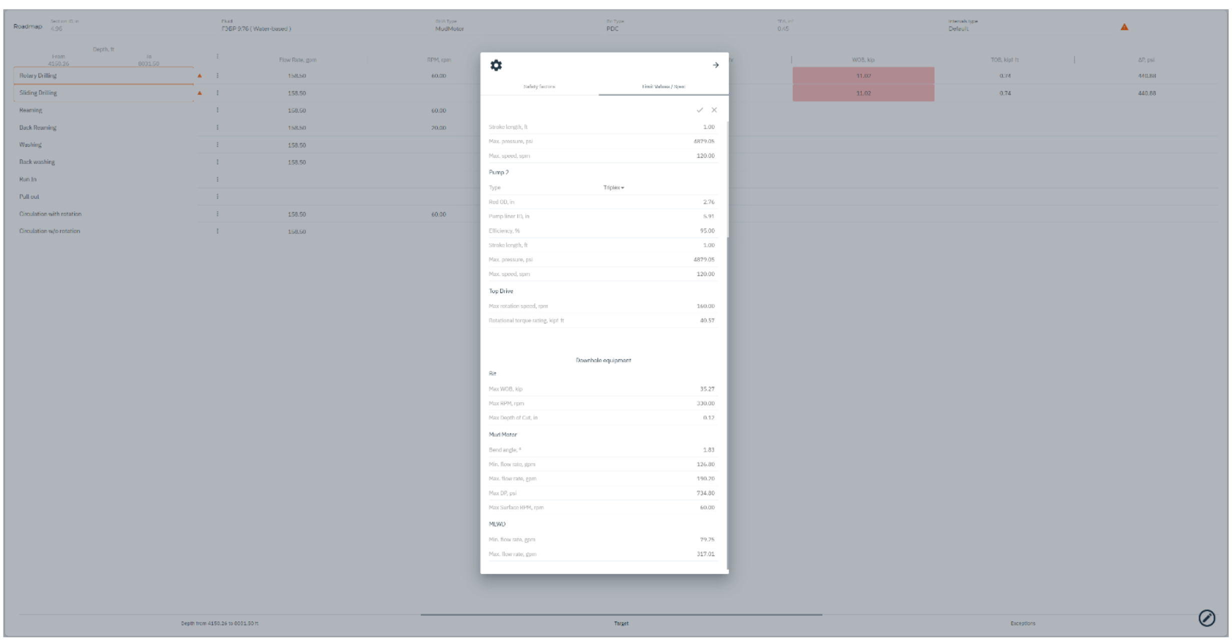

Figure 5—Equipment specs input table

To support trajectory control and collision avoidance, the ecosystem includes an automated slide sheet log that captures detailed information about drilling modes, operational parameters, and MWD survey data. It calculates the projected "blind zone" to the bottom of the well and uses this to assess alignment with the planned trajectory. Actual survey data is monitored in real time to detect deviations, and the system uses the minimum curvature method to forecast the corrective path needed to return to plan, while also accounting for potential collision risks. Based on this forecast, the system automatically defines the optimal sequence of drilling modes and recommends the Toolface Orientation (TFO) required for maintaining trajectory control during slide drilling (Fig.6).

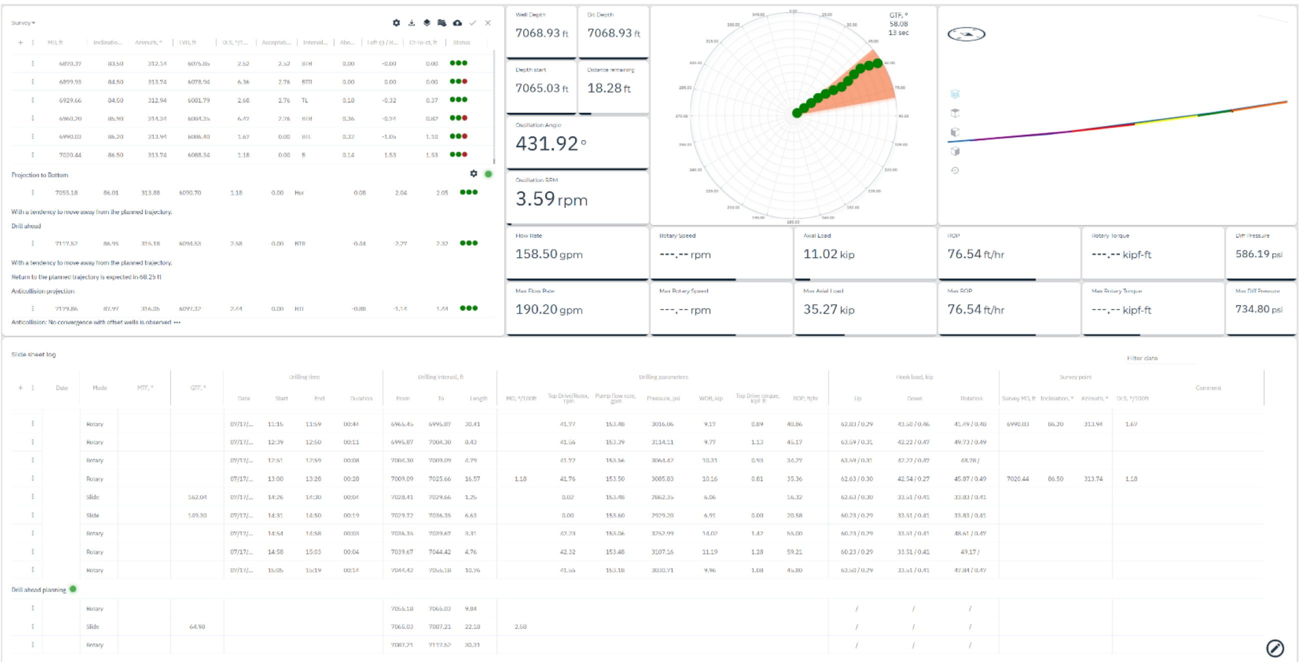

Figure 6—Ecosystem «autoslide» module interface

Figure 7—«Autoslide» calculation results description

Using this data, the AutoSlide module generates a detailed drilling plan for each stand. It specifies the required lengths to be drilled in rotary and slide modes and determines the TFO necessary for effective slide drilling. All setpoints defined by the roadmap and the automated slide sheet are transferred to the Operation Queue Builder.

Based on the initial operation—whether rotary or slide—the Operation Queue Builder generates a corresponding sequence of preparation steps. These may include operations such as reaming, running in hole (RIH), and rotating on bottom (ROB). The user also has the option to define the number of reciprocation cycles to be performed. Additionally, the Operation Queue Builder supports auto-hold positioning and automatic Bottom Hole Assembly (BHA) orientation when drilling with a Positive Displacement Motor (PDM).

The generated operation sequence and corresponding setpoints are displayed on the driller's HMI panel (Fig.8), ensuring full visibility during execution. The directional drilling (DD) engineer can also access the same data through the ecosystem monitoring module, maintaining alignment between surface personnel and downhole operations.

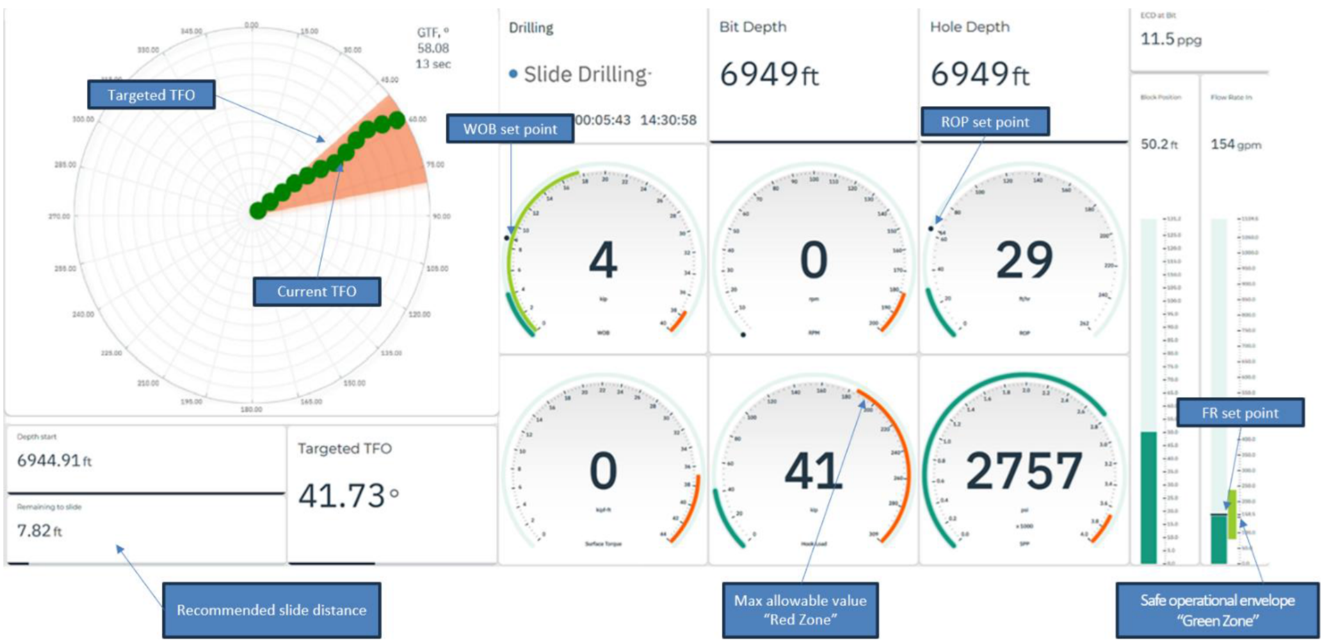

Figure 8—Driller's HMI

Once the setpoints are reviewed and approved, they are transmitted to the PLC. Based on the approved plan and the configured operation type, the autodriller switches to automated mode and executes the drilling sequence accordingly—managing pumps, top drive, and drawworks.

Upon completing the stand, the autodriller brings the top drive to a position approximately 3 feet above the rotary table to initiate a connection and automatically switches to manual mode, handing over connection execution to the rig crew.

To ensure optimal drillstring movement, the digital twin continuously monitors mechanical loads during reaming operations and identifies signs of overpull or slack-off. If such anomalies are detected, the system adjusts the setpoints and instructs the PLC to initiate additional reciprocation within the affected interval. This real-time response reduces friction and restores effective movement. Furthermore, during slide drilling, if weight transfer issues are detected, the torque and drag (T&D) model recommends pipe rocking strategies. It provides guidance on the optimal rotational speed and number of rotations required to improve weight transfer and maintain drilling efficiency.

Results and Observations

The implementation of the ecosystem integrated with the autodriller system has demonstrated a number of positive outcomes:

Improved Rate of Penetration (ROP)

Field practice has shown a significant increase in mechanical drilling speed (ROP) on wells where the ecosystem was deployed in conjunction with the advanced autodriller system. This improvement is attributed to the system's ability to quickly adapt drilling parameters to changing downhole conditions, minimizing downtime and optimizing the drilling process.

The example below illustrates the difference in well construction time for two wells—X-1 and X-2—with similar hole section diameters (11 ⅝″ and 8 ½″), recorded before and after the implementation of the ecosystem managing the autodriller.

- Well X-1 was drilled prior to the deployment of the ecosystem.
- Well X-2 was drilled with the ecosystem fully integrated with the autodriller unit.

Despite the similarity in well profile and formation type, Well X-2 demonstrated a significant reduction in overall construction time, primarily due to:

- Improved Rate of Penetration (ROP) through optimized, real-time control of drilling parameters.
- Reduced non-productive time (NPT) due to better event detection and automated mitigation strategies.
- Enhanced connection practices and minimized flat time, supported by seamless automation of slips-to-slips operations.

This comparison clearly highlights the operational efficiency gains enabled by the integration of a digital ecosystem with autodriller control.

Implemented systems collaboration on well X-2 resulted in drill time savings while drilling 11-⅝″ section (8h) and while drilling 8-½″ section (13.5h), in total 21.5h (Fig. 10).

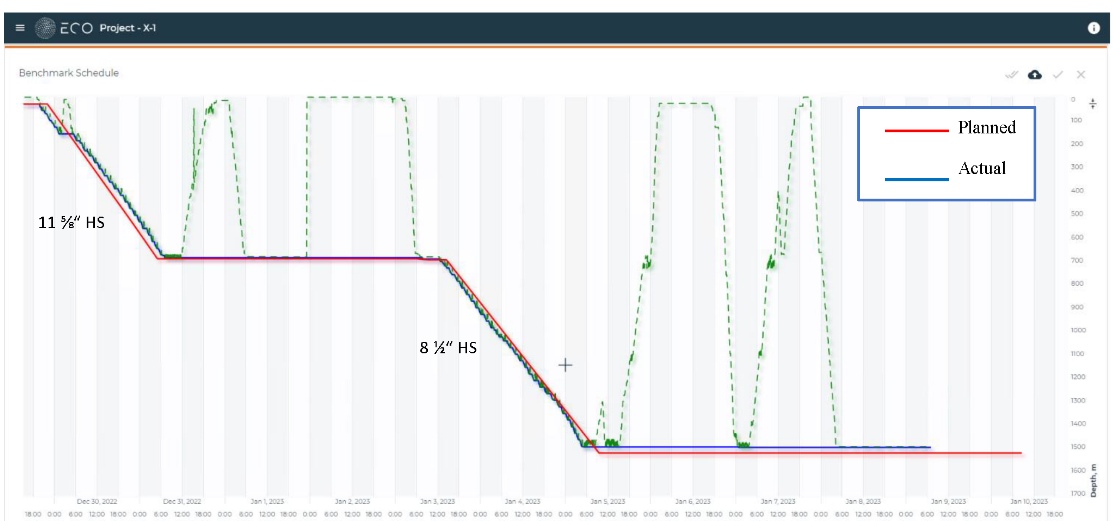

Figure 9—Well X-1 DvsD graph before the ecosystem implementation

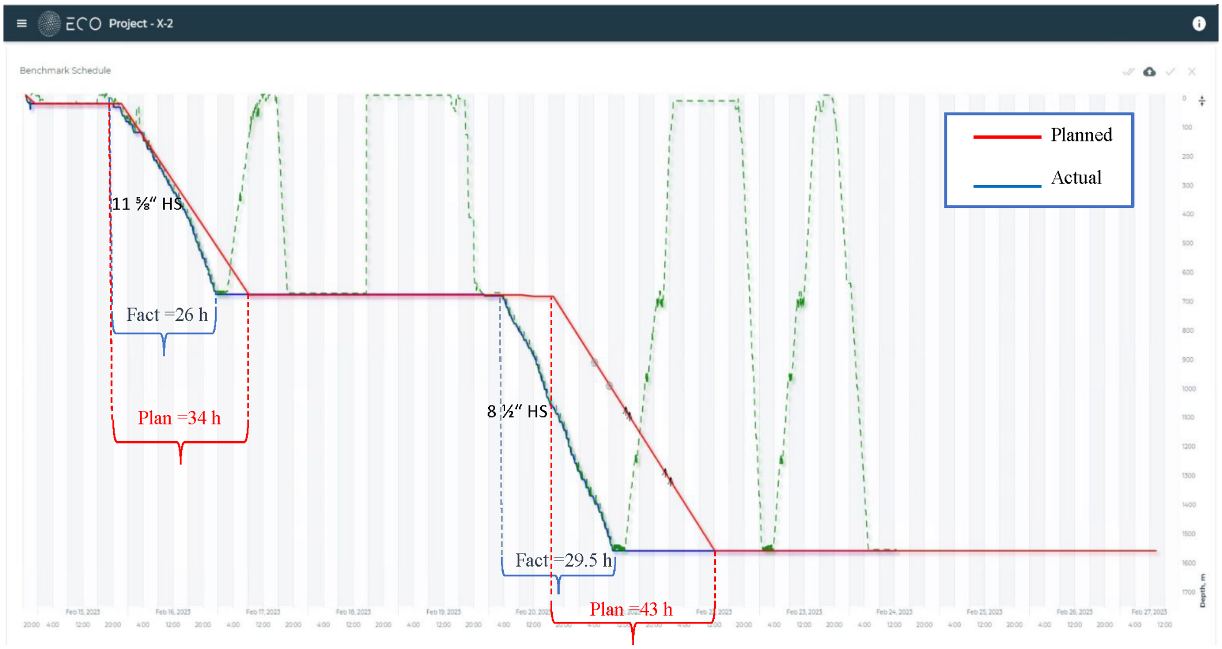

Figure 10—Well X-2 DvsD graph after the ecosystem implementation

Reduction in Non-Productive Time (NPT)

The use of the ecosystem led to a substantial decrease in incidents that typically cause non-productive time, such as:

- Stuck pipe and losses due to exiting safe technological margins.
- Borehole packing caused by insufficient hole cleaning.
- Helical buckling and drillstring wear due to excessive WOB.
- Planned trajectory violations due to inadequate directional drilling planning.
- Uncontrolled borehole tortuosity resulting from exceeding dogleg severity (DLS)

Transition to Safer and More Efficient Drilling. The dynamic digital twin incorporated in the ecosystem has not only improved operational efficiency but has also significantly enhanced drilling safety. Through proactive data analysis and real-time updates, the system helps prevent many potential hazards before they escalate into serious incidents.

Applicability in Narrow Operating Windows. In environments with narrow operational windows—where accessibility and safety are critical—the technology has proven to be highly effective. Its dynamic control capabilities and flexibility in decision-making ensure robust performance under changing operational conditions.

Conclusion

The integration of autonomous drilling systems—particularly advanced autodrillers—with a real-time digital ecosystem enhanced by dynamic digital twin technology marks a transformative step forward in the automation of well construction. This combined solution moves beyond conventional control approaches by enabling real-time, context-aware decision-making that accounts for surface equipment, downhole dynamics, and formation behavior simultaneously.

By continuously updating the digital twin with high-frequency data from surface and downhole sensors, and applying advanced AI-driven analytics, the system enables precise control of drilling parameters, early detection of operational anomalies, and predictive adjustments to prevent drilling dysfunctions. This leads to measurable improvements in drilling performance and wellbore quality.

Field results confirm the benefits of this integration. Wells drilled with the ecosystem-autodriller combination showed significantly improved mechanical drilling speed (ROP), reduced Non-Productive Time (NPT), and enhanced connection practices. A direct comparison between wells X-1 and X-2—with similar well designs—demonstrated a total time savings of 21.5 hours, achieved through more efficient drilling of 11⅝″ and 8½″ sections. These results underscore the value of real-time automation in reducing flat time and accelerating overall well construction.

In environments with narrow operational windows, where precision and adaptability are critical, the ecosystem's ability to dynamically manage drilling modes, maintain trajectory control, and respond autonomously to evolving downhole conditions ensures safe and robust execution. The seamless coordination of drilling, reaming, tripping, and connection phases through automated sequencing minimizes manual intervention and supports consistent adherence to the drilling roadmap.

Ultimately, the fusion of autonomous drilling technology with an intelligent digital ecosystem and a dynamic digital twin enables a new level of operational efficiency, safety, and consistency. As the oil and gas industry continues to advance toward more sustainable and cost-effective drilling operations, this integrated approach will be a key enabler in meeting both present and future challenges.

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